

SMARTBRANCH 360

Branch Network Design and Assurance

Cisco Packet Tracer | VLAN Segmentation | DHCP | DNS | ACL | NAT | SSH | Python Assurance

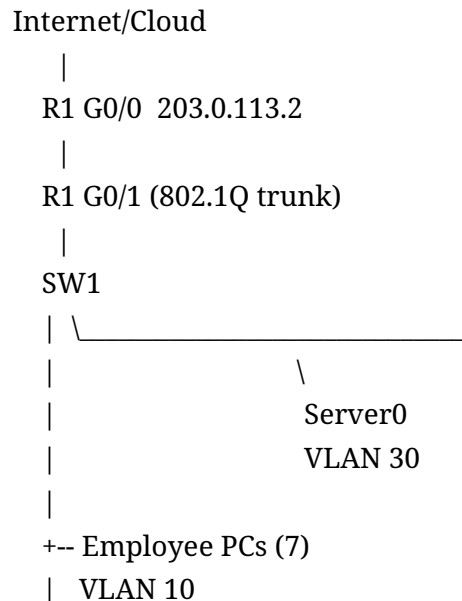
Item	Design choice
Router	R1 Cisco 2911
Switches	SW1 and SW2 Cisco 2960-24TT
Wireless	AP1 with WPA2-PSK, SSID SmartBranch-Guest
Addressing	10.10.10.0/24, 10.10.20.0/24, 10.10.30.0/24, 10.10.99.0/24
Management	VLAN 99; SSH restricted to management subnet
Assurance	Python checker compares requirements.yaml with saved show outputs

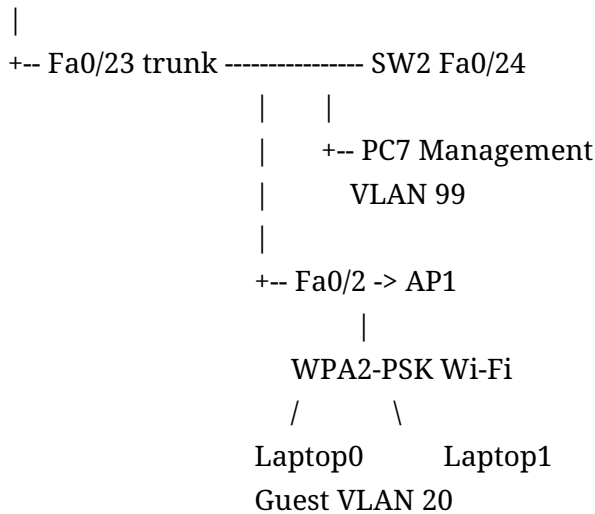
1. Objectives

- Build a segmented branch network for Employee, Guest, Server, and Management traffic.
- Provide DHCP per client VLAN, a static server address, DNS service, inter-VLAN routing, Guest isolation, NAT configuration, and secure SSH administration.
- Demonstrate repeatable troubleshooting using intentional faults and a Python network assurance checker.

2. Topology and Device Placement

Logical topology





3. VLAN and IP Plan

VLAN	Name	Subnet	Gateway	Devices
10	EMPLOYEE	10.10.10.0/24	10.10.10.1	7 Employee PCs
20	GUEST	10.10.20.0/24	10.10.20.1	AP1 + 2 Guest laptops
30	SERVER	10.10.30.0/24	10.10.30.1	Server0 = 10.10.30.10 (static)
99	MANAGEMENT	10.10.99.0/24	10.10.99.1	PC7 + switch management SVIs

4. Port Mapping

Device	Port	Connection	Role
SW1	Fa0/1	Server0	Access VLAN 30
SW1	Fa0/2-Fa0/8	7 Employee PCs	Access VLAN 10
SW1	Fa0/23	SW2 Fa0/24	802.1Q trunk
SW1	Fa0/24	R1 G0/1	802.1Q trunk
SW2	Fa0/1	PC7	Access VLAN 99
SW2	Fa0/2	AP1	Access VLAN 20

5. Core Services and Routing

- Router-on-a-stick: R1 G0/1 provides subinterfaces G0/1.10, G0/1.20, G0/1.30, and G0/1.99.
- DHCP: R1 provides separate pools for Employee, Guest, and Management. Reserved management addresses include 10.10.99.1-10.10.99.10.
- Server: Server0 uses static 10.10.30.10/24 with gateway 10.10.30.1 and provides DNS service.
- NAT: R1 marks G0/1.10, G0/1.20, G0/1.30, and G0/1.99 as inside and G0/0 as outside. PAT uses ACL 1 and G0/0.
- WAN: R1 G0/0 uses 203.0.113.2/30 in the lab design. Actual end-to-end Internet reachability through Packet Tracer Cloud must be verified separately before the final recording.

6. Wireless Design

Setting	Value
SSID	SmartBranch-Guest
Authentication	WPA2-PSK
Encryption	AES
Use	Guest-only wireless access
Switch port	SW2 Fa0/2, access VLAN 20

7. Security Rules

- Guest VLAN 20 is permitted to reach the intended outside network but is denied access to Server VLAN 30 and Management VLAN 99.
- Guest DHCP and DNS traffic are explicitly permitted before the isolation rules.
- Only the Management subnet 10.10.99.0/24 is permitted on device VTY lines for SSH administration.
- SSH uses local authentication and RSA keys. Telnet is disabled on the VTY lines.

Guest ACL sequence:

1. permit DHCP client requests
2. permit UDP DNS to 10.10.30.10
3. permit TCP DNS to 10.10.30.10
4. deny Guest -> Server VLAN
5. deny Guest -> Management VLAN
6. permit Guest -> any

8. Python Assurance Tool

The assurance checker uses a YAML requirement file plus saved Cisco show-command outputs. It currently validates:

- VLAN/IP plan
- Trunk VLANs
- R1 subinterface IPs and up/up state
- DHCP pool network ranges
- Guest ACL rules
- NAT inside/outside and overload rule

Example PASS output:

```
[PASS] All required VLANs are present.  
[PASS] GigabitEthernet0/1.20: 10.10.20.1 | up/up  
[PASS] DHCP pool GUEST: network 10.10.20.0/24.  
[PASS] Guest ACL contains all required security rules.  
[PASS] NAT overload rule is configured.
```

9. Fault and Troubleshooting Method

1. Start from a known-good baseline and save device configurations.
2. Introduce one fault at a time without reloading devices.
3. Capture the relevant show output and user-visible symptom.
4. Run the Python checker to identify the likely configuration problem.
5. Apply the smallest fix, verify with ping/show commands, and restore the healthy state.
6. Repeat for five documented fault scenarios.

10. Troubleshooting Notes

Issue observed	Diagnosis	Resolution
Guest laptop received 169.254.x.x	Guest DHCP path was broken; trunk/ACL path was inspected.	Restore VLAN 20 on trunks and permit DHCP in Guest ACL.
SW1 reported duplicate 10.10.99.2	Management DHCP range overlapped static SW1 management IP.	Exclude 10.10.99.1-10.10.99.10 from DHCP; Management PC received 10.10.99.11.
SW1 SSH session closed	SW1 had no RSA keys; SSH was disabled.	Set hostname/domain and generate 1024-bit RSA keys; enable SSH v2.
Router subinterface IP error	VLAN 20/30 subinterface IPs were initially swapped and G0/1.30 lacked encapsulation.	Configure dot1Q 20/30 first, then assign 10.10.20.1 and 10.10.30.1.

11. Final Verification Checklist

- Employee PCs obtain 10.10.10.x through DHCP.
- Guest laptops obtain 10.10.20.x through WPA2-PSK Wi-Fi and DHCP.
- Management PC obtains an address outside the reserved static switch IP range.
- Server0 is static at 10.10.30.10.
- R1 subinterfaces are up/up with correct gateways.
- All required trunk VLANs are allowed and active.
- Guest cannot reach Server/Management while DNS is permitted.
- Management PC can SSH to R1, SW1, and SW2.
- Python checker returns PASS for the saved healthy baseline.
- Five fault scenarios and the 5-10 minute demo are prepared.